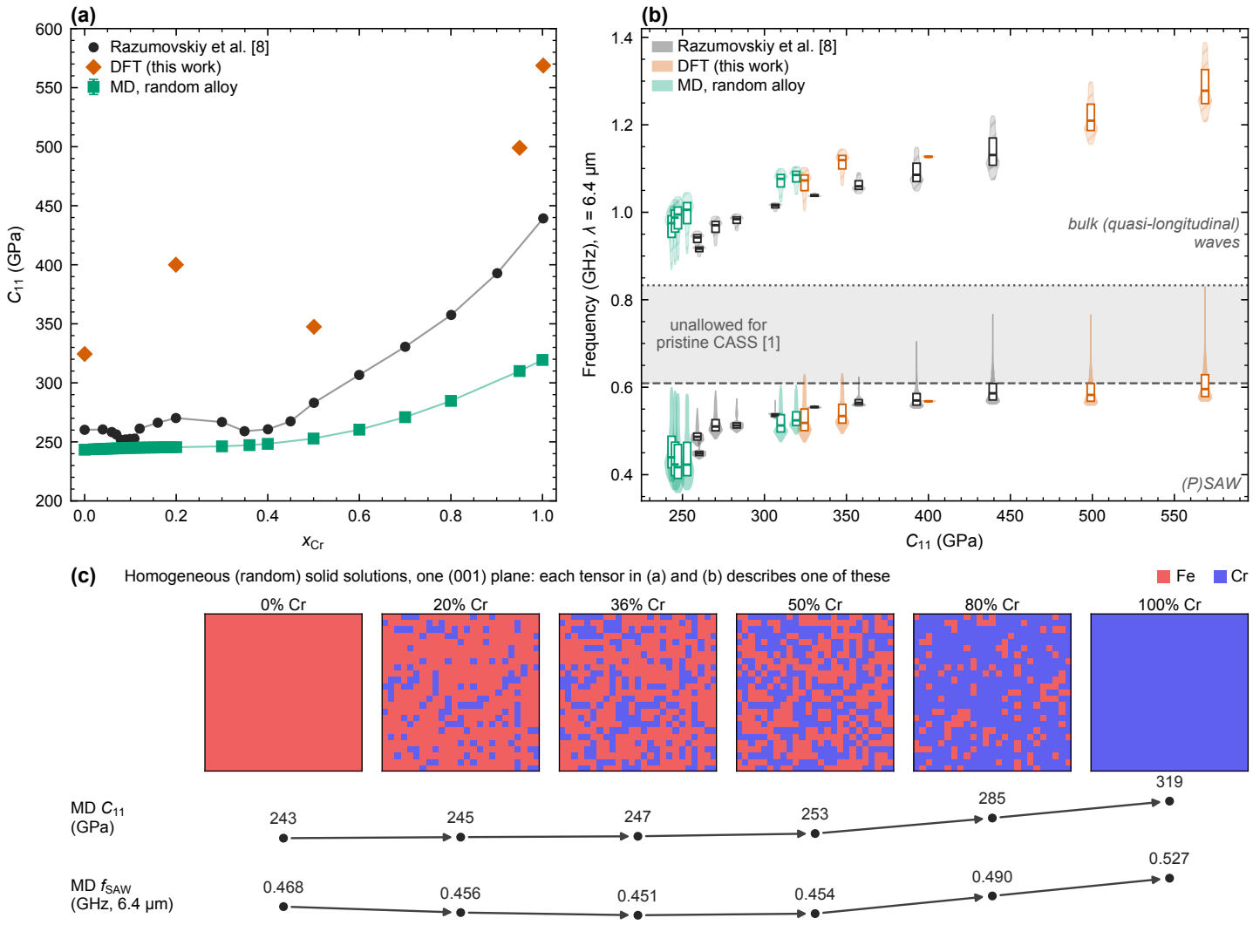

(a)
Razumovskiy et al. [8]
DFT (this work)
MD, random alloy
C_{11} (GPa)
x_{Cr}
(b)
Razumovskiy et al. [8]
DFT (this work)
MD, random alloy
Frequency (GHz), λ = 6.4 μm
bulk (quasi-longitudinal) waves
unallowed for pristine CASS [1]
(P)SAW
C_{11} (GPa)
(c)
Homogeneous (random) solid solutions, one (001) plane: each tensor in (a) and (b) describes one of these
Fe
Cr
0% Cr
20% Cr
36% Cr
50% Cr
80% Cr
100% Cr
MD C_{11} (GPa)
243
245
247
253
285
319
MD f_{SAW} (GHz, 6.4 μm)
0.468
0.456
0.451
0.454
0.490
0.527